

Bayesian inference projects

Advanced statistics for physics analysis

May 29, 2026

Each project is intended for a group of at most 3 students and should be feasible in a few days. The emphasis is on Bayesian modeling, parameter inference, model comparison, and prediction. Students should use either a hand-coded MCMC sampler, JAGS, Stan, or another transparent Bayesian workflow implemented in R. For every project, the final report should contain the likelihood, priors, posterior summaries, convergence diagnostics, posterior predictive checks, and the requested model comparison.

P01 – Earthquakes in Italy: Gutenberg–Richter law and spatial clustering

Field: Geophysics

Scientific question. Are the frequency–magnitude relation and the spatial distribution of strong earthquakes in and around Italy consistent with a single homogeneous seismic region, or do different regions have different parameters?

Data. Download earthquake events from the USGS Earthquake Catalog using the following CSV query:

- <https://earthquake.usgs.gov/fdsnws/event/1/query?format=csv&starttime=1923-01-01&endtime=2025-12-31&minmagnitude=5&minlatitude=35&maxlatitude=48&minlongitude=5&maxlongitude=20&orderby=time-asc>
- Documentation: <https://earthquake.usgs.gov/fdsnws/event/1/>

Models.

- Model 1: one Gutenberg–Richter model for the whole region, with

$$\log_{10} N(M_i \geq M) = a - bM,$$

$N(M_i \geq M)$ being the number of events with magnitude equal to or larger than M .

- Model 2: separate Gutenberg–Richter slopes for northern, central and southern Italy, using latitude bands chosen before fitting.
- For the temporal component, model yearly counts above a fixed magnitude threshold using a Poisson likelihood with either constant rate or time-varying rate.

Required output.

1. Map the epicentres and plot the empirical cumulative magnitude distribution.
2. Infer the posterior distribution of the Gutenberg–Richter parameter b .
3. Compare the one-region and three-region models.
4. Compute posterior predictive distributions for the number of earthquakes with $M \geq 5$ in a future 10-year interval.

References.

- B. Gutenberg and C. F. Richter, “Frequency of earthquakes in California”, *Bulletin of the Seismological Society of America* 34, 185–188 (1944).
 - USGS Earthquake Hazards Program, Earthquake Catalog API: <https://earthquake.usgs.gov/fdsnws/event/1/>
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P02 – ARPAV temperature trends in Veneto region

Field: Meteorology and Climate

Scientific question. Is there evidence for a systematic trend in temperature at meteorological stations in Veneto?

Data. Use the ARPAV open-data archive “Principali variabili meteorologiche”, distributed as ZIP files containing CSV data for individual stations:

- <https://www.arpa.veneto.it/dati-ambientali/open-data/clima/principali-variabili-meteorologiche>
- Alternative CKAN page: <https://opendata.arpa.veneto.it/dataset/principali-variabili-meteorologiche>

Use 3 stations (for example, Auronzo di Cadore, Castelfranco Veneto, and Porto Tolle).

Models.

- Model 1: constant mean annual temperature for each station.
- Model 2: linear trend in time for each station,

$$T_{s,t} = T_s^0 + \dot{T}_s t + \epsilon_{s,t}$$

where s labels the station, and ϵ is a Gaussian noise with variance σ_s^2 .

Required output.

1. Plot monthly and annual averages of minimum, mean and maximum temperature.
2. Infer station-specific parameters and their credible intervals.
3. Compare constant and linear-trend models.
4. Produce posterior predictions for the next 10 years for the annual mean temperature.

References.

- ARPAV, Open Data Meteo e Clima: <https://www.arpa.veneto.it/dati-ambientali>
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P03 – COVID-19 Vaccine efficacy from clinical trial counts

Field: Biomedical Statistics

Scientific question. What is the posterior uncertainty on vaccine efficacy when efficacy is inferred from case counts in randomized vaccine trials?

Data. Use official FDA clinical-review documents for COVID-19 vaccines. Use with the Pfizer–BioNTech/Comirnaty BLA clinical review memorandum, which reports trial counts for vaccine and placebo groups:

- FDA Comirnaty clinical review memorandum: <https://www.fda.gov/media/152256/download>
- FDA Comirnaty page: <https://www.fda.gov/vaccines-blood-biologics/comirnaty>

Models.

- Model 1: independent binomial risks in vaccine and placebo arms,

$$Y_V \sim \text{Binomial}(n_V, p_V), \quad Y_P \sim \text{Binomial}(n_P, p_P),$$

where Y_V (Y_P) is the number of cases in the vaccine (placebo) group made up of n_V (n_P) people, p_V (p_P) is the the risks of disease in the vaccine (placebo) arms. The vaccine efficacy is defined as

$$\text{VE} = 1 - \frac{p_V}{p_P}.$$

- Model 2: logistic model with a treatment indicator,

$$Y_i \sim \text{Binomial}(n_i, p_i), \quad \text{logit}(p_i) = \alpha + \beta x_i,$$

where $x_i = 1$ for the vaccine arm and $x_i = 0$ for the placebo arm. In this model

$$p_P = \text{logit}^{-1}(\alpha), \quad p_V = \text{logit}^{-1}(\alpha + \beta),$$

and therefore

$$\text{VE} = 1 - \frac{\text{logit}^{-1}(\alpha + \beta)}{\text{logit}^{-1}(\alpha)}.$$

The parameter β is the log odds ratio.

Required output.

1. Extract the numbers of participants and COVID-19 cases in vaccine and placebo arms from the FDA document.
2. Infer the posterior distribution of VE and report a 95% credible interval.
3. Compare the binomial-risk model and the logistic-treatment model.
4. Compute the posterior predictive distribution of cases in a hypothetical trial with 10^4 participants per arm.

References.

- FDA, *BLA Clinical Review Memorandum: COMIRNATY*, <https://www.fda.gov/media/152256/download>
 - F. P. Polack et al., “Safety and efficacy of the BNT162b2 mRNA Covid-19 vaccine”, *New England Journal of Medicine* 383, 2603–2615 (2020).
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P04 – Exoplanet radius distribution: one or two populations?

Field: Astrophysics

Scientific question. Is the observed distribution of exoplanet radii compatible with a single population, or is there evidence for two broad populations?

Data. Use the NASA Exoplanet Archive. The following TAP query downloads confirmed planets with measured radius and orbital period:

- https://exoplanetarchive.ipac.caltech.edu/TAP/sync?query=select+pl_name,host_name,pl_rade,pl_orbper,disc_year,discoverymethod+from+pscomppars+where+pl_rade+is+not+null+and+pl_orbper+is+not+null&format=csv
- TAP documentation: <https://exoplanetarchive.ipac.caltech.edu/docs/TAP/usingTAP.html>

Models.

- Model 1: single log-normal distribution for planet radius,

$$\log R_i \sim \mathcal{N}(\mu, \sigma^2).$$

- Model 2: two-component mixture of log-normal distributions,

$$\log R_i \sim w\mathcal{N}(\mu_1, \sigma_1^2) + (1 - w)\mathcal{N}(\mu_2, \sigma_2^2).$$

Required output.

1. Restrict the dataset to planets with $R < 7R_\oplus$.
2. Infer the posterior parameters of the one- and two-population models.
3. Compare the two models using approximate Bayes factor (or another clearly stated Bayesian model-comparison criterion).
4. Report the posterior probability that a randomly selected planet has radius below $2R_\oplus$.

References.

- NASA Exoplanet Archive, TAP service: <https://exoplanetarchive.ipac.caltech.edu/docs/TAP/usingTAP.html>
- B. J. Fulton et al., “The California-Kepler Survey. III. A gap in the radius distribution of small planets”, *Astronomical Journal* 154, 109 (2017).

P05 – Global CO_2 climate trend: linear or accelerating?

Field: Climate Physics

Scientific question. Mauna Loa is the world's premier site for tracking atmospheric CO_2 concentration and its variation due to anthropogenic activities. Is CO_2 concentration better described by a linear increase or by an accelerating trend?

Data. Use NOAA Global Monitoring Laboratory monthly CO_2 averages:

- Dataset page: <https://gml.noaa.gov/data/dataset.php?item=mlo-co2-observatory-monthly>
- Direct text file: https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.txt

Models. Consider CO_2 data averaged over 1 year, to exclude seasonal variations.

- Model 1: linear trend,

$$y_t = \alpha + \beta t + \epsilon_t.$$

- Model 2: quadratic trend,

$$y_t = \alpha + \beta t + \gamma t^2 + \epsilon_t.$$

Required output.

1. Plot the monthly CO_2 series removing missing values coded in the data file, as well as the annual average.
2. Infer the trend parameters, including the acceleration parameter γ .
3. Compare the linear and quadratic trend models using approximate Bayes factor (or another clearly stated Bayesian model-comparison criterion).
4. Predict the annual CO_2 concentration for the next 10 years with credible bands.

References.

- NOAA Global Monitoring Laboratory, Mauna Loa CO_2 monthly averages: <https://gml.noaa.gov/data/dataset.php?item=mlo-co2-observatory-monthly>
 - IPCC Sixth Assessment Report, Working Group I (The Physical Science Basis), Chapter 5 https://www.ipcc.ch/report/ar6/wg1/downloads/report/IPCC_AR6_WGI_Chapter05.pdf.
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P06 – Arrhenius or super-Arrhenius relaxation time of glasses?

Field: Statistical physics of disordered systems

Scientific question. Rapidly lowering the temperature below the melting point can cause a material to vitrify, namely, to transform into an amorphous state with huge viscosity. Does the relaxation time (that is proportional to the viscosity) of a three-dimensional glass-forming liquid grow according to an Arrhenius law, or is there evidence for super-Arrhenius behavior as temperature is lowered?

Data. Use dataset obtained by numerical simulations given in the Zenodo archive hereafter. It contains the file `3d_Relaxation_Time.csv`, which reports the relaxation time as a function of inverse temperature for a 3D soft-sphere model of glass-forming liquids:

- <https://zenodo.org/records/15178992>
- Paper: <https://doi.org/10.1103/PhysRevE.111.045416>

The dataset includes relaxation times for several Monte Carlo dynamics. You may analyze one dynamics, or compare results of two dynamics.

Models. Let $x_i = 1/T_i$, and let τ_i be the relaxation time.

- Model 1: Arrhenius law,

$$\log \tau_i = A + E x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

This corresponds to

$$\tau(T) = \tau_0 \exp(E/T).$$

- Model 2: Vogel–Fulcher–Tammann super-Arrhenius law,

$$\log \tau_i = A + \frac{B}{T_i - T_0} + \epsilon_i, \quad T_0 < \min_i T_i.$$

The CSV file reports statistical uncertainties on the measured relaxation times. In the main analysis, treat these uncertainties as known measurement errors. Work with $y_i = \log \tau_i$, propagating the reported uncertainty as $\sigma_{y_i} = \frac{\delta \tau_i}{\tau_i}$. Optionally, use a Gaussian likelihood that includes an additional intrinsic scatter term

$$y_i \sim \mathcal{N}\left(f(T_i; \theta), \sigma_{y_i}^2 + \sigma_{\text{int}}^2\right),$$

and infer σ_{int} .

Required output.

1. Plot $\log \tau$ as a function of $1/T$ for the 3D dataset.
2. Report posterior distributions and credible intervals for the model parameters.
3. Compare models using posterior predictive checks.
4. Produce posterior predictive bands for $\tau(T)$ over the observed temperature range.
5. Predict the value of the relaxation time at a temperature $T_{\text{new}} = 0.8 \min_i T_i$.

References.

- Y. Nishikawa, F. Ghimenti, L. Berthier, and F. van Wijland, “Irreversible swap algorithms for soft sphere glasses”, *Physical Review E* **111**, 045416 (2025).
- L. Berthier and G. Biroli “Theoretical perspective on the glass transition and amorphous materials” *Rev. Mod. Phys.* **83**, 587 (2011).

P07 – Cosmic-ray flux spectrum: single or broken power law?

Field: Astroparticle Physics

Scientific question. Is the high-energy cosmic-ray proton spectrum compatible with a single power law, or is there evidence for a spectral break/hardening?

Data. Use the AMS-02 proton flux data from Table I of the 2015 proton-flux publication. The table gives the proton flux as a function of rigidity R from 1 GV to 1.8 TV, together with statistical and systematic uncertainties. Rigidity, being defined as the momentum per unit charge, is proportional to energy for energies much larger than the rest mass.

- Direct CSV download: <https://ams02-d10.web.cern.ch/sites/default/files/publication/201501/table-i.csv>
- AMS-02 publication page: <https://ams02.space/publications>

Use rigidity R , flux Φ , statistical uncertainty, and total systematic uncertainty. Combine the uncertainties in quadrature,

$$\sigma_{\Phi} = \sqrt{\sigma_{\text{stat}}^2 + \sigma_{\text{syst}}^2}.$$

To avoid low-rigidity effects such as solar modulation, restrict the fit to $R > 45$ GV.

Models. Fit the models in log-log space with Gaussian errors $\sigma_{\log \Phi} = \frac{\sigma_{\Phi}}{\Phi}$.

- Model 1: single power law,

$$\Phi(R) = AR^{-\gamma}.$$

- Model 2: broken power law with break rigidity R_b ,

$$\Phi(R) = AR^{-\gamma_1} \quad (R < R_b),$$

$$\Phi(R) = AR_b^{\gamma_2 - \gamma_1} R^{-\gamma_2} \quad (R \geq R_b).$$

Required output.

1. Plot the measured proton flux with error bars on log-log axes.
2. Infer the spectral index γ for the single power law.
3. Infer γ_1 , γ_2 , and R_b for the broken power law.
4. Compare the single and broken power-law models.
5. Predict the proton flux at three rigidities inside the fitted range and one rigidity just outside its upper limit.

References.

- AMS Collaboration, “Precision measurement of the proton flux in primary cosmic rays from rigidity 1 GV to 1.8 TV with the Alpha Magnetic Spectrometer on the International Space Station”, *Physical Review Letters* 114, 171103 (2015).
- Cosmic-Ray Data Base (CRDB): <https://lpsc.in2p3.fr/crdb/>.

P08 – Allometric scaling in animals: is Kleiber’s law universal?

Field: Biology and Biophysics

Scientific question. Many biological quantities obey approximate power-law scaling relations. Does animal basal metabolic rate scale with body mass according to Kleiber’s law, $B \propto M^{3/4}$, or is the exponent different? Is a single exponent adequate for all animals, or do different taxonomic groups have different scaling exponents?

Data. Use the AnimalTraits database, a curated open database of terrestrial animal traits including body mass, metabolic rate and brain size:

- Dataset page: <https://animaltraits.org/>
- Zenodo record: <https://zenodo.org/records/6468938>
- Direct data file: <https://zenodo.org/records/6468938/files/observations.csv>
- Column documentation: <https://zenodo.org/records/6468938/files/column-documentation.csv>

Select N observations with both body mass and whole-body metabolic rate available. Retain only rows for which `metabolic rate - method` is `basal metabolic rate`. Use the standardized `metabolic rate` column, in watts, as the response variable. Analyze N animals together, or restrict the analysis to two taxonomic groups, birds and mammals.

Models. Let M_i be body mass and B_i metabolic rate. Work in log–log space:

$$x_i = \log M_i, \quad y_i = \log B_i.$$

- Model 1: Kleiber fixed-exponent model,

$$y_i = \alpha + \frac{3}{4}x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- Model 2: free allometric exponent,

$$y_i = \alpha + \beta x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2).$$

- Model 3: group-dependent scaling,

$$y_i = \alpha_{g[i]} + \beta_{g[i]}x_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2),$$

where $g[i]$ denotes the taxonomic group of observation i .

Required output.

1. Plot metabolic rate versus body mass on log–log axes, distinguishing the main taxonomic groups.
2. Infer the posterior distribution of the allometric exponent β .
3. Compute the posterior probability $P(\beta > 3/4 \mid \text{data})$ and report whether Kleiber’s exponent is inside the 95% credible interval.
4. Compare the fixed-exponent, free-exponent and group-dependent models.
5. Produce posterior predictive bands for metabolic rate over the observed body-mass range.
6. Predict the metabolic rate of an animal with mass $M = 1$ kg, including posterior uncertainty.

References.

- M. Kleiber, *The Fire of Life: An Introduction to Animal Energetics*, Wiley (1961).
- M. E. Herberstein et al., “AnimalTraits – a curated animal trait database for body mass, metabolic rate and brain size”, *Scientific Data* 9, 265 (2022).